CS3281 / CS5281

Process Scheduling

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Spring 2026

**Some lecture slides borrowed and adapted from "Operating Systems: Three Easy Pieces"*

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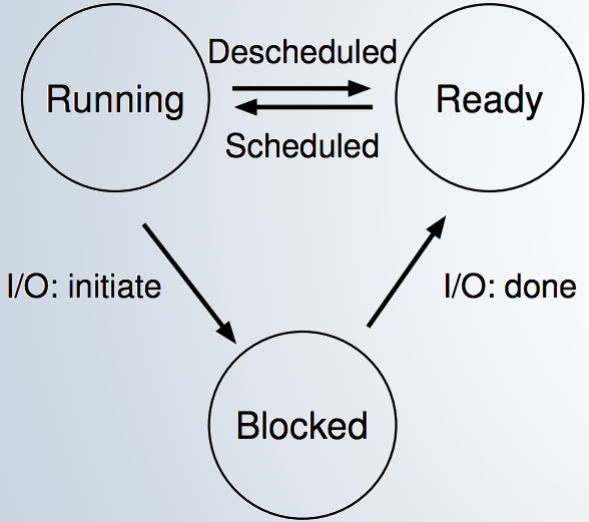
Overview

- The scheduler is the part of the OS kernel responsible for deciding which process runs and for how long

- There are many types of scheduling algorithms
  - Which one is “best” depends on many things, such as the expected workload and metrics

Processes don't run all the time

The three basic process states:



- OS ***schedules*** processes
  - Decides which of many competing processes to run.
- A ***blocked*** process is not ready to run and is waiting on I/O
- I/O means input/output – anything other than computing.
  - For example, reading/writing disk, sending network packet, waiting for keystroke, condvar/semaphore!
  - While waiting for results, the OS **blocks** the process, waiting to do more computation until the result is ready

More about I/O

- All “interesting” programs do some I/O
- What should the scheduler do when a process is waiting for I/O?
  - Run another process instead! It’s a bad idea to have a process spin while waiting for I/O
- Example from the book:

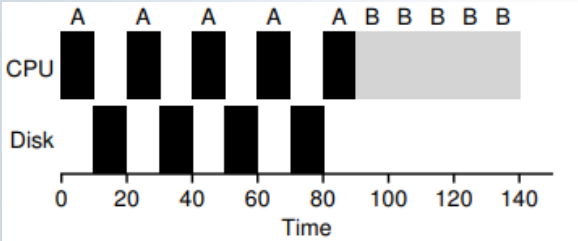


Figure 7.8: **Poor Use Of Resources**

vs

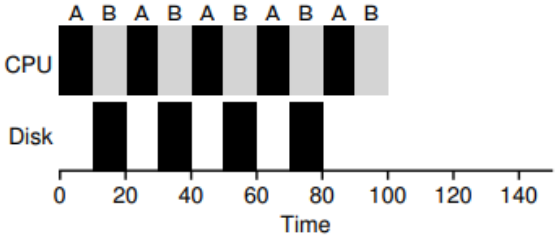
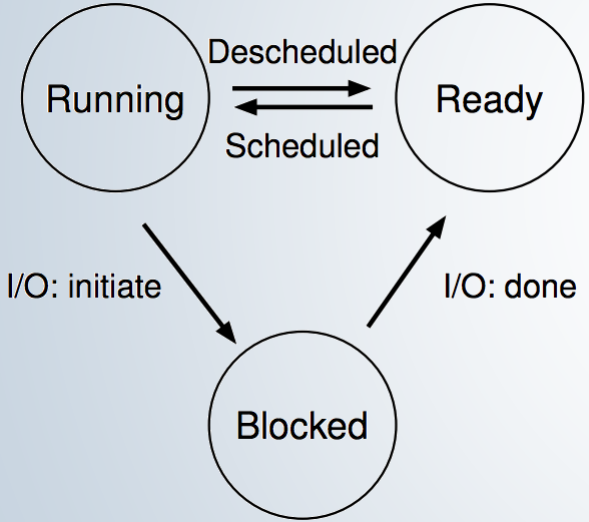


Figure 7.9: **Overlap Allows Better Use Of Resources**

Multiprogramming processes

The three basic process states:



- Even with a single processor, the OS can provide the illusion of many processes running simultaneously
  - And also use this opportunity to get more useful work done

- When one process is Blocked, OS can schedule a different process that is Ready
- OS can also swap between various Ready processes so they all make progress

Scheduling

- We know that multiple processes will be sharing the CPU
  - Possibly multiple threads in each process
  - Possibly multiple cores in the CPU

- Scheduling is creating a *policy* for sharing the CPU
  - Which process/thread is chosen to run, and when?
  - When (if ever) does the OS change which process is running?

Scheduling

- A scheduling algorithm is responsible for deciding which process to run
- Some simple (impractical) scheduling algorithms (chapter 7 in your book). These algorithms schedule the *batch* (non-interactive, aka. No I/O) processes
  - First In First Out (FIFO): run each process to completion in the order they arrive. If multiple processes have the same arrival time, a process is selected randomly
    - Problems: what if a process runs “indefinitely”? What if a long running process happens to arrive first?
  - Shortest Job First (SJF): select the process that will run for the shortest time. If multiple processes have the same time, select a process with the earliest arrival time
    - Problem: we don’t know how long a process will run ahead of time!
  - Shortest Time-to-Completion First (STCF): *preempt* the currently running process if another process has a shorter time to completion. If multiple processes have the same time, select a process with the earliest arrival time
    - Problem: long running processes will be *starving* if short running processes keep arriving

Scheduling Metrics

- Metrics can be used to measure how good a scheduler is
- Examples:
  - Turnaround time: Time of completion - Time of arrival
    - A scheduler with poor turnaround time might be bad for CPU bound processes
    - Maximize the number of processes completed per unit of time
  - Response time: Time of first response - Time of arrival
    - Time between pressing a key and seeing a character on screen
    - A scheduler with a poor response time will feel “laggy” to interactive users
  - Throughput: Number of processes completed over a specific time interval.
  - Waiting time: The total amount of time that a process waits on getting scheduled on a CPU.

More Realistic Scheduling: Round-Robin

- Round-Robin (RR) scheduling: instead of running a process to completion, run a process for a *time slice* (or time *quantum*), *preempt* the process, and context switch to the next process in the run queue; do this repeatedly
  - In other words: everybody gets a turn eventually!
  - Processes are in a single queue. The priority of a process is based on its position in the queue. (The process at the head of the queue has the highest priority.) When the time slice expires, the process moves to the tail of queue
  - RR is used in many real-world schedulers.

Exercise

Schedule these processes with FIFO, SJF, SJCF, and RR algorithms

- The time slice is 1 unit of time for RR
- If two or more processes are eligible to run at an instant, the one with the lowest number is scheduled

Process	Runtime	Arrival time
P1	10	0
P2	1	3
P3	2	1
P4	1	2
P5	5	0

Exercise

FIFO

0	10	15	17	18	19
P1	P5	P3	P4	P2	

SJF

0	5	6	7	9	19
P5	P2	P4	P3	P1	

SJCF

0	1	3	4	5	9	19
P5	P3	P2	P4	P5	P1	

RR

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	19
P1	P5	P1	P3	P5	P4	P1	P2	P3	P5	P1	P5	P1	P5	P1	

CPU Bound vs. I/O Bound

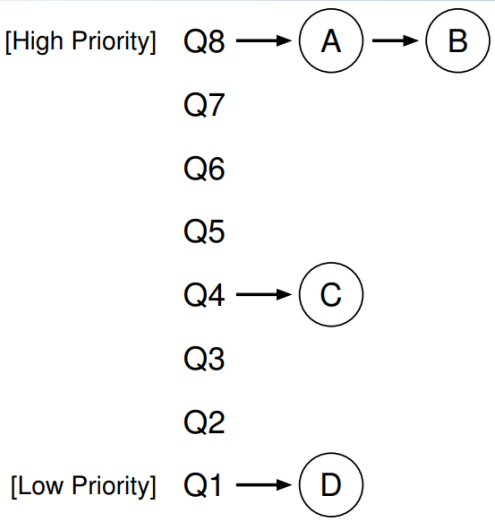
- Processes are often *CPU bound* or *I/O bound*
  - CPU bound: when they run, they use the CPU continuously
    - Examples: MATLAB simulations, training a machine learning model, video encoding
  - I/O bound: when they run, they spend most of their time waiting for I/O
    - Examples: text editors, PowerPoint, shell
- Processes can switch between being I/O bound and CPU bound (and vice versa)
  - Example: when you do a “spell check” on a document, the editor goes from being I/O bound to CPU bound

Multi-Level Feedback Queues (MLFQ)

- MLFQ is a scheduling algorithm that tries to do two things:
  - Optimize turnaround time
    - This can be done by running shorter processes first, but we don't know how long a process will run!
  - Minimize response time
    - This makes the system feel responsive to interactive users
    - Round-robin is good for response time, but is bad for turnaround time
- MLFQ is cool because it addresses these two goals without knowing anything about the running times beforehand

MLFQ Basic Rules

- MLFQ has:
  - A number of distinct queues, each at a different priority level
  - At any given time, a process that is ready to run is on a single queue
  - Processes with higher priority are run first; processes with equal priority are run RR
- Rule 1: priority(A) > priority(B) => A runs
- Rule 2: priority(A) == priority(B), A and B run in RR



Changing Priority

- Rule 3: When process enters system, it is placed at highest priority (top queue)
- Rule 4a: If a process uses up an entire time allotment while running, its priority is reduced (it moves down to the tail of a queue with a lower priority)
  - The lower priority queues have longer time allotment
- Rule 4b: If a process gives up the CPU before its time allotment is up, it stays at the same priority level

Example of Moving through Queues

- Over time, a long running process will move to the bottom queue
- Example:

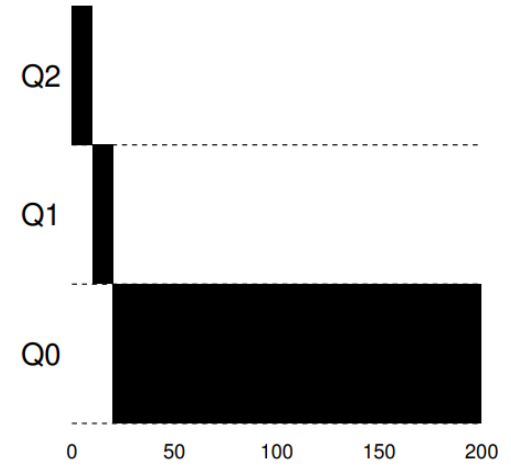


Figure 8.2: **Long-running Job Over Time**

More Examples

- Figure 8.3: a long running process and short-running job
- Figure 8.4: a long running process and an I/O bound process

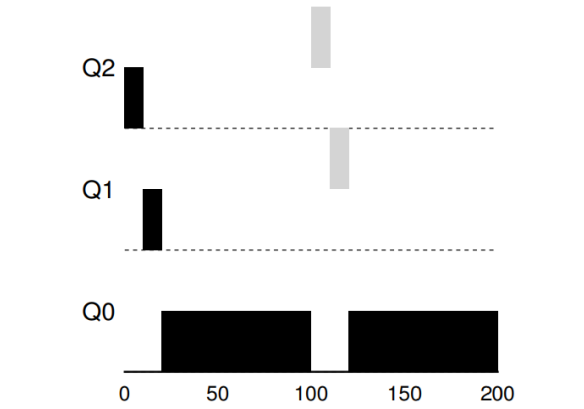


Figure 8.3: **Along Came An Interactive Job**

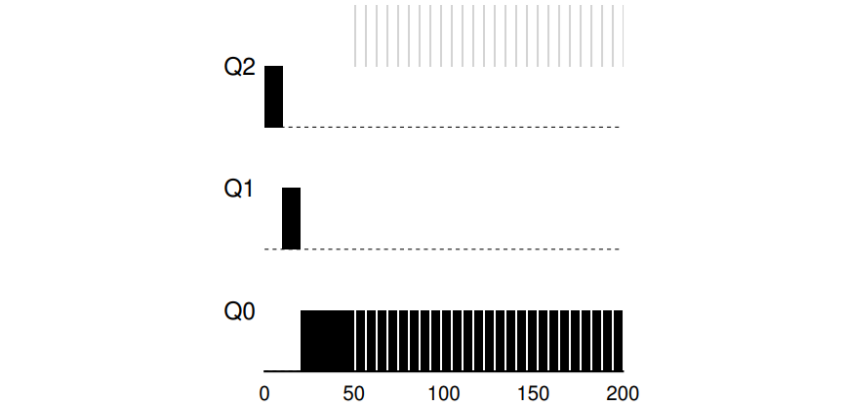


Figure 8.4: **A Mixed I/O-intensive and CPU-intensive Workload**

Problems

- There are two problems with the design so far:
  - Processes have no way to move back up in priority (starvation)
    - Starvation is the indefinite delay to execute a process when the other processes are allowed to proceed
  - Processes can “game” the scheduler by giving up the CPU right before using the quantum

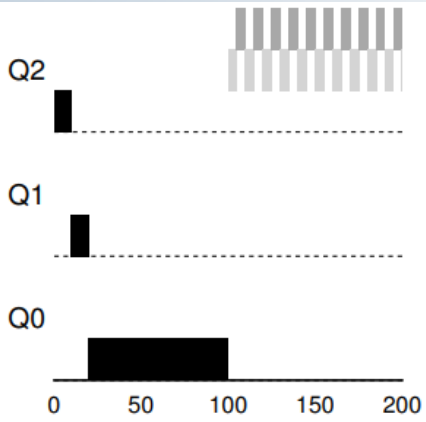


Figure 8.5: **Without (Left) and With (Right) Priority Boost**

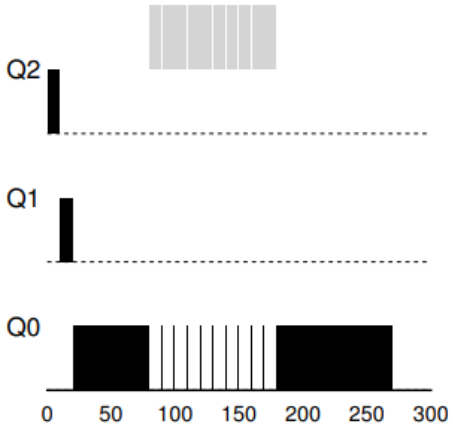


Figure 8.6: **Without (Left) and With (Right) Gaming Tolerance**

Addressing the Problems

- We can keep processes from “gaming” the scheduler by moving them down a queue after they exhaust their cumulative time slice at a given level
- We can move processes up in priority periodically to keep them from starving
  - As an extension: we could “weight” processes so that some get to move up faster (but not too fast!)

Summary of MLFQ Rules

- Rule 1: priority(A) > priority(B) => A runs
- Rule 2: priority(A) == priority(B), A and B run in RR
- Rule 3: when process enters system, it is placed at highest priority (top queue)
- Rule 4: once a process uses its allotment at a given level, its priority is reduced (it moves down to the tail of a queue with a lower priority)
- Rule 5: after some time period *S*, move all the jobs in the system to the topmost queue

Exercise

Consider these processes with the arrive times and runtimes.

Suppose you have 3 queues: the top queue uses RR with time slice of 8 time units, the middle queue uses RR with time slice of 16 time units and the bottom queue uses RR with time slice of 32 time units to schedule processes. How do you schedule these processes?

Process	Runtime	Arrival time
P1	36	0
P2	20	16
P3	12	20

Exercise

	0	8	16	24	32	40	52	56	68
Q2	P1		P2	P3					
Q1		P1			P1	P2	P3		
Q0								P1	

Multitasking: Preemptive Operating Systems

- Linux, Windows, macOS, iOS, are preemptive, multitasking OSes
  - Preemptive: the scheduler decides when a process should stop and another should start
  - Multitasking: it can simultaneously interleave execution of more than one process
- On a single-processor machine, this makes it seem like multiple processes are running concurrently
- On a multiprocessor (multicore) machine this allows multiple processes to run in parallel on different processors (cores)
- The *timeslice* (or quantum) is the amount of time a process runs before being preempted

Review: Preemption

- What triggers the preemption of the currently running process?

- Case 1: returning from interrupt handler
  - Example: the timer interrupt occurs
  - OS handles this interrupt by executing the timer interrupt handler
  - Timer interrupt handler checks how long current process has been running
  - If current process has run longer than its allotted time slice, the scheduler is invoked
  - The scheduler saves the context of the current process and selects a new process to run
- Case 2: process makes a system call
  - Recall: process generates an intentional exception called a trap (e.g., `int 80h, syscall`)
  - OS executes a trap handler (to handle the system call)
  - When exiting the system call, the OS checks if it should invoke the scheduler
  - The scheduler, if run, checks for higher priority processes

High-Level Concept: Lottery Scheduling

- Simple idea: every so often, hold a lottery to determine which process should run next
  - Processes that should run more often get more *tickets*, and thus more chances to win the lottery
  - Described in chapter 9 in the book
- Example:
  - Two processes, A and B
  - 100 tickets total. A has 75, B has 25
  - Pick a random winning ticket to see who gets to run next
  - A should get the CPU ~75% of the time, B should get it ~25% of the time
- Need a good random number generator, a data structure for tickets, and the total number of tickets!

Fairness

- Each job should get a "fair" share of the processor
- Fair means different things
  - Equal CPU time for all runnable processes
  - CPU time proportional to priority or weight
  - No process waits forever (no starvation)
  - Similar processes are treated similarly
  - Each process gets a turn within a reasonable time
  - Interactive or short jobs are not unfairly delayed
  - Low-priority jobs still make progress
  - Resources are shared in a predictable, balanced way
- Scheduler should be fair with regards to the goals of the system it runs on

Completely Fair Scheduler Scenario

- The CFS in Linux is similar to lottery scheduling
- Consider the following scenario
  - Two tasks: a text editor (I/O bound) and a MATLAB simulation (CPU bound)
  - The text editor needs to respond to key presses quickly
  - MATLAB needs CPU time to perform a simulation
- In the ideal case:
  - Give a larger proportion of CPU time to text editor
    - But not because it needs it! Because we want it to have time the moment it needs it.
  - Allow the text editor to preempt MATLAB as soon as input is available (i.e., when a key is pressed)
    - This will make it responsive and give good interactive performance

Virtual Runtime

- So how is the timeslice calculated?
  - Too low: increases “fairness” but also increases overhead (due to context switching)
  - Too high: decreases overhead but also decreases fairness
- Timeslice uses a “magic value” (empirically determined) called `sched_latency` (typical value = 48ms) to calculate the timeslice
  - Timeslice = sched_latency / # of processes
- Nice value: parameter of a process controllable by user
  - Goes from -20 to +19; default of 0;
    - Negative implies higher priority (you are “less nice”)

Completely Fair Scheduler Scenario (cont.)

- Divides CPU time evenly among processes
  - Instead of a “fixed” timeslice, CFS calculates how long each process should run as a function of the total number of runnable processes
    - Use the nice value to weight this proportion of processor a process receives
    - If all nice values are equal: all processes get an equal proportion of processor time
  - Uses a simple counting technique known as virtual runtime (vruntime)
    - Lower vruntime => a process hasn’t had its “fair share”

Timeslice

- Use the nice value to "weight" the timeslice

$$\texttt{time_slice}_k = \frac{\texttt{weight}_k}{\sum_{n=0}^{n-1} \texttt{weight}_i} \cdot \texttt{sched_latency} \qquad (9.1)$$

- Also scale the `vruntime`:

prio_to_weight[20]

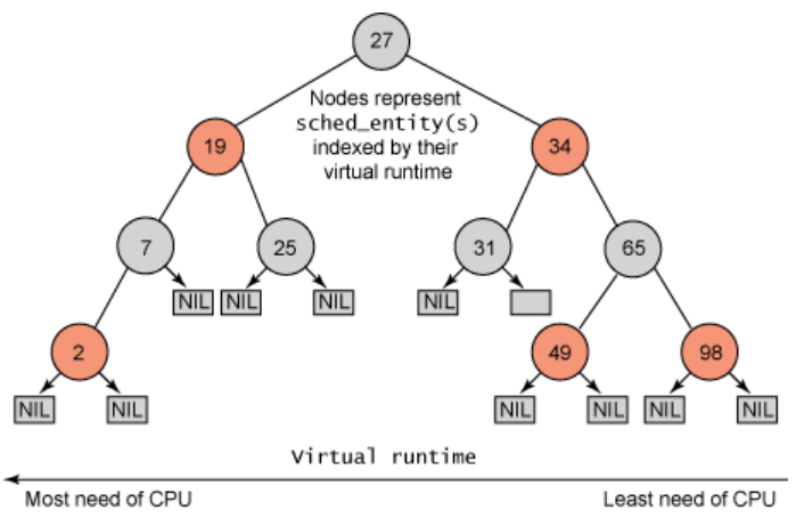
$$\texttt{vruntime}_i = \texttt{vruntime}_i + \frac{\texttt{weight}_0}{\texttt{weight}_i} \cdot \texttt{runtime}_i \qquad (9.2)$$

- Table of weights:

```
static const int prio_to_weight[40] = {
 /* -20 */     88761,     71755,     56483,     46273,     36291,
 /* -15 */     29154,     23254,     18705,     14949,     11916,
 /* -10 */      9548,      7620,      6100,      4904,      3906,
 /*  -5 */      3121,      2501,      1991,      1586,      1277,
 /*   0 */      1024,       820,       655,       526,       423,
 /*   5 */       335,       272,       215,       172,       137,
 /*  10 */       110,        87,        70,        56,        45,
 /*  15 */        36,        29,        23,        18,        15,
};
```

Virtual Runtime (cont.)

- But what if there are too many processes?
  - Define a minimum timeslice value; never use timeslice lower than this
- But how do we give processes “priority”?
  - In CFS: give them a larger share of the CPU

Virtual Runtime (cont.)

- As a process runs, it accumulates vruntime
- When scheduler needs to pick a new process, it picks the process with the lowest vruntime